

Vertical sorting in bed forms: Flume experiments with a natural and a trimodal sediment mixture

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Received 26 November 2001; revised 27 June 2002; accepted 27 June 2002; published 5 February 2003.

[1] Two sets of flume experiments were conducted to examine grain size selective transport and vertical sorting in conditions with migrating bed forms and bed load transport. In the two sets of experiments we used a sediment mixture from the river Rhine and a trimodal mixture, respectively. The vertical sorting profiles showed a downward coarsening trend within the bed forms, and in some experiments an essentially immobile coarse bed layer formed underneath the migrating bed forms. Three mechanisms contribute to the formation of such a coarse bed layer: (1) the avalanching process at the lee face, (2) conditions of partial transport in which a relatively large amount of coarse material does not participate in the transport process, and (3) the winnowing of fines from the trough surface and subsurface. The experiments show that vertical sorting fluxes not only occur through net degradation or aggradation but also through the migration of bed forms and through the variability in trough elevations. This is contradictory to the way vertical sorting processes are modeled in most existing sediment continuity models for nonuniform sediment. The present study is therefore also a plea for modifying existing sediment continuity models to account for vertical sorting processes other than through net aggradation or degradation.

INDEX TERMS: 1824 Hydrology: Geomorphology (1625); 1815 Hydrology: Erosion and sedimentation; **KEYWORDS:** vertical sorting, flume experiments, nonuniform sediment, bed forms, sediment transport, mobile armoring

Citation: Blom, A., J. S. Ribberink, and H. J. de Vriend, Vertical sorting in bed forms: Flume experiments with a natural and a trimodal sediment mixture, *Water Resour. Res.*, 39(2), 1025, doi:10.1029/2001WR001088, 2003.

1. Introduction

[2] Morphological changes of the river bed result from gradients in the transport rate, which is expressed by the Exner equation of sediment continuity or the mass balance equation. In cases where the bed material consists of multiple size fractions, gradients in the transport of size fractions will result in (1) a change in the composition of the bed surface and/or (2) net degradation or aggradation of the river bed. A further complication in the mass balance is the variation of the bed composition over depth. Vertical sediment fluxes within the bed result in a certain organization of particles over depth (the vertical sorting), which influences the rate and composition of the sediment transport and the bed forms. As a result, it may cause changes in bed roughness, river morphology, and water levels during floods.

[3] For a plane bed, the vertical sorting is typically characterized by a coarse top layer (mobile pavement or armor layer), whereas, in dune conditions, the bed material shows a downward coarsening trend [Bagnold, 1941; Allen,

1965] (see Figure 1). This downward coarsening arises from the size selective sorting processes at the bed form lee face. Single coarse grains roll further down the lee face than finer grains, since (1) their velocity rolling down the lee face is larger than for finer grains, (2) they experience less friction than finer grains, and (3) they experience less support to find a resting place [Zanke, 1976]. When a mass of grains (an avalanche) moves down the lee face, another mechanism plays a dominant role in the lee face sorting process. Bagnold [1941] and Allen [1965] observed that the fines filter down to the interface over which the avalanche is moving and the coarse grains are worked up to the surface of the avalanche. As such, the finer grains move more slowly and become lodged more easily, whereas the coarser grains overtake the finer ones and move more easily to the toe of the lee face. For an overview of research in this field and recent flume experiments on sediment sorting down a delta face, see Kleinhans [2002].

[4] When flow conditions are well above the threshold for incipient motion for all grain sizes in the mixture, the coarse grains in the lower parts of the bed forms participate in the transport process. This was the case in the experiments by Ribberink [1987], who used a mixture of two well sorted sand fractions ($d_{50,1} = 0.78$ mm, $d_{50,2} = 1.29$ mm). When a mixture

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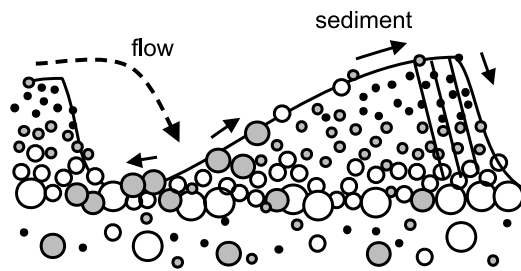


Figure 1. Downward coarsening of dune material and a gathering of coarse material underneath migrating dunes. Revised from Zanke [1976].

is more widely graded and a relatively large amount of the coarse grains do not or hardly participate in the transport process, an essentially immobile coarse layer may form below the migrating bed forms [Zanke, 1976; Willis, 1988; Wilcock and Southard, 1989; Klaassen, 1990; Blom et al., 2000; Kleinhans, 2002]. This is illustrated in Figure 1. Wilcock and Southard [1989] called the coarse layer “a partial static armor, wherein some individual grains become essentially immobile even though other grains in the same fraction remain in transport” and remark that “size dependent vertical sorting can serve to concentrate these coarser grains on the bed surface over which the bed forms move”. At higher flows, they observed that the coarse layer was eliminated by scour in the lee of the dunes. This was confirmed by Klaassen [1990]: “The occurrence of bedforms initiated the breaking up of the armor layer. This was caused by the turbulence generated in the wake of the bedforms.”

[5] Another mechanism that contributes to the formation of a coarse bed layer is the winnowing or washing out of fines from the trough surface and subsurface [Willis, 1988; Wilcock and Southard, 1989; Klaassen, 1990], although it has never been directly measured. Wilcock and Southard [1989] remark that “A coarse surface layer and bed forms coexist over a finite range of τ . When both are present, the coarse surface layer does not constitute the entire bed surface that is visible at any time, but is exposed only in the bed form troughs. The coarse surface layer is the surface on which, or through which, size-dependent exchange of grains occurs. As such, the coarse surface layer plays a role in determining the grain size distribution of the transported sediment and therefore of the bedforms.” Based on observations in the flume experiments by Klaassen [1986], Klaassen [1990, p. 8] mentions the process of winnowing: “The material from which these bedforms were made of originated from underneath the armor layer and was sucked through this layer from underneath slightly unstable individual particles.” Klaassen found that the coarse layer lowered through the process of winnowing. This was confirmed by Willis [1988]: “With the passage of subsequent dune troughs, the gravel particles sank further until finally the gravel constituted a new layer at or even slightly below the deepest trough penetration of the dunes.”

[6] In numerical models for rivers with nonuniform sediment, the interaction among grain size selective transport, bed level changes, and the vertical sorting profile is described by so-called sediment continuity models, e.g., the active layer model by Hirano [1971]. However, the Hirano active layer model and its variants do not adequately

describe the vertical sorting processes within the bed and their influence on the sediment transport processes. For example, they do not account for the downward coarsening within bed forms, the formation of coarse bed layers, and the influence of coarse layers upon sediment entrainment. Moreover, in most sediment continuity models, vertical sediment fluxes only occur when there is a change in the average bed level, whereas previous research has shown that this is not required for sediment exchange fluxes to occur [Ribberink, 1987; Wilcock and Southard, 1989; Klaassen, 1992]. The two layer model by Ribberink [1987] was the first exception, by taking into account vertical sediment exchange processes due to the variability in bed form trough elevations. A recent and important step forward was made by Parker et al. [2000], who developed a new type of framework for sediment continuity without discrete bed layers. The framework is based on the probability density of bed surface elevations and describes (the time evolution of) the vertical sorting profile and the vertical sediment fluxes continuously over depth. The present paper does not go into the translation of the present results into improvements to existing sediment continuity models, yet it helps to better understand their shortcomings and it is a plea for improving them.

[7] As far as known to the authors, up to now, the experiments by Ribberink [1987] and Klaassen [1992] are the only sets of flume experiments with bed forms and nonuniform sediment in which the interaction among size selective transport, bed level changes, the probability density of bed surface elevations, and the vertical sorting profile was studied in detail. The motive of the present study was the need for more data on vertical sorting processes within bed forms under well controlled laboratory conditions to further develop the above sediment continuity models. This paper presents two new sets of flume experiments that were aimed at conditions with bed forms, partial transport with coarse bed layers, and bed load transport. Partial transport is here defined as the situation in which the transported material is finer than the bed material, as the coarse grains are either transported in a smaller proportion than their presence in the bed or not at all [Wilcock and McArde, 1993]. In the first set of experiments (1997/1998), we used a sediment mixture from the river Waal, which is a branch of the river Rhine. In this part of the Rhine it has been observed that coarse bed layers form during flood events at the base of the large dunes [Kleinhans, 2002]. To facilitate the analysis of the vertical sorting processes, we used an artificial mixture of three well sorted grain sizes (a trimodal mixture) in the second set of experiments (1999/2000). During each experiment, the sediment was recirculated and uniform flow conditions were maintained. In this way, the average bed level remained constant, so that we could focus on the interaction between only size selective transport, the probability density of bed surface elevations, and the vertical sorting profile. The present experiments differ from those by Ribberink [1987] in their more widely graded sediment mixtures. The results of some of the experiments by Klaassen [1992] ($d_{50} = 0.6$ mm, $\sigma_g = 2.3$) indicate that the coarse material in the lower part of the bed forms had formed a more or less immobile coarse layer.

[8] The focus of the present paper is on the experiments with the trimodal mixture, as measurements were performed

more often and in greater detail. Besides, in the experiments with the Rhine mixture, the objective of having conditions with only bed load transport was not fully met, as the fines were transported as suspended load. Yet, the experiments with the Rhine mixture are included in the paper, as they clearly show that the processes observed for the artificial trimodal mixture also occur for a natural mixture.

2. Experimental Setup

2.1. Sand Flume Facility

[9] The two sets of experiments were conducted in the Sand Flume facility at WL Delft Hydraulics in the Netherlands [Blom and Kleinhans, 1999; Blom, 2000]. The length, width, and height of the flume's measurement section are 50 m, 1.5 m, and 1.0 m, respectively. The discharge is controlled by water pumps and a weir at the upstream end of the flume. The downstream water level is controlled by a tailgate at the downstream end of the flume. The sediment is recirculated. Just upstream of the tailgate, the transported sediment is caught in a sediment trap. From there, it is pumped to a hydro-cyclone at the upstream end of the flume. There, the sediment settles to the bottom of the hydro-cyclone. Its remaining volume is filled with water. When the submerged weight of the sediment in the hydro-cyclone reaches a specific value (which occurs when the total weight of the hydro-cyclone reaches a specified value), the sediment is released back into the flume. By recording the moments of these sediment deposits, the total load transport (bed load and suspended load) can be calculated. To determine the grain-size distribution of the total load transport, samples from the sediment deposits were taken regularly. In the second set of experiments (with the trimodal mixture), the width of the flume was reduced from 1.5 m to 1.0 m. This was done to reduce the bed form adaptation length (i.e. the length from the upstream sediment inlet over which the bed forms increase in height), as previous experiments showed that the bed form adaptation length decreases with decreasing width [Ribberink, 1987].

2.2. Experimental Program

[10] In the first set of experiments, we used a sediment mixture dredged from the river Waal, a branch of the river Rhine, near Pannerdensch Kop in the Netherlands ($D_{10} =$

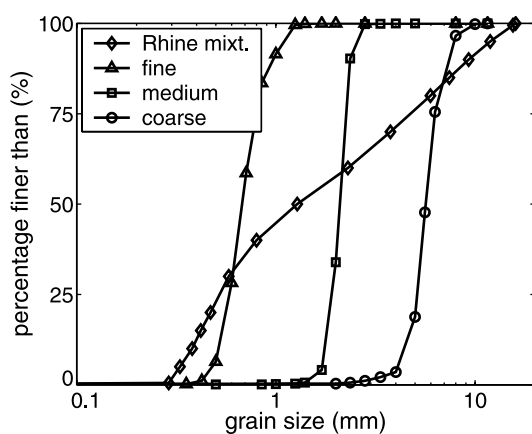


Figure 2. Grain size distributions in the experiments with the Rhine mixture and the trimodal mixture.

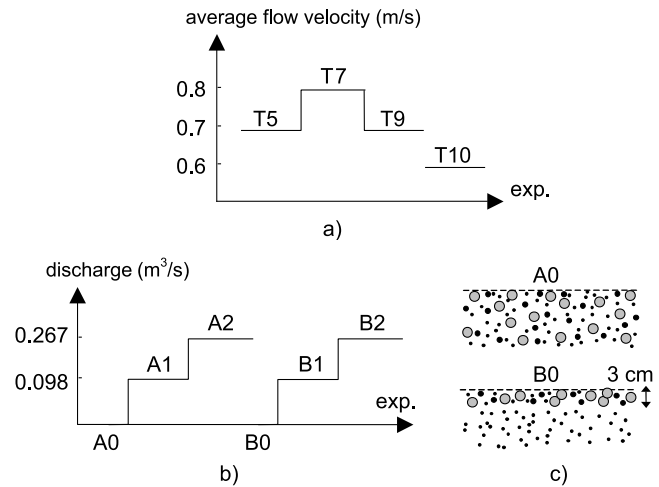


Figure 3. (a) Average flow velocities in experiments with Rhine mixture, (b) discharges in the experiments with the trimodal mixture, and (c) initial sorting profiles A0 and B0.

0.38 mm, $D_{50} = 1.3$ mm, and $D_{90} = 9.3$ mm; see Figure 2). Sediment particles coarser than 16 mm (the D_{95} of the original mixture) were removed from the original mixture, as these were too coarse to be transported through the sediment recirculation system. The sediment mixture used in the second set of experiments consisted of three well sorted size fractions: fine $D_{50} = 0.68$ mm, medium $D_{50} = 2.1$ mm, and coarse $D_{50} = 5.7$ mm (Figure 2).

[11] In each experiment, the transported sediment was recirculated and uniform flow was maintained, so that the average bed level remained constant (no net degradation or aggradation). To this end, the slope of the water level was kept equal to the originally installed bed slope. During an experiment the bed forms developed toward new equilibrium dimensions, which induced changes in bed roughness. Thus, to maintain uniform flow the boundary conditions needed to be adjusted slightly. Each experiment was run until the flow and transport reached an equilibrium stage in which the bed roughness, sediment transport rate and composition, and bed form pattern and dimensions varied about a stable mean.

[12] In each of the experiments with the Rhine mixture, the average flow velocity was held constant. To maintain both uniform flow and a constant flow velocity, both the downstream water level and the discharge were adjusted slightly. Equilibrium was reached in four experiments with bed forms: T5, T7, T9, and T10 (Figure 3a). Experiments T5 and T10 started with a new fully mixed sediment bed. Experiments T7 and T9 started from the final stages of the preceding experiments, i.e., T5 and T7, respectively. In experiment T9, the flow velocity was reduced to the same level as in T5, so that T5, T7, and T9 can be considered as a flood hydrograph.

[13] Four experiments were conducted with the trimodal mixture: A1, A2, B1, and B2 (Figure 3b). To control uniform flow, now for practical reasons the discharge was kept constant and only the downstream water level was adjusted. In experiments A1 and B1, the discharge was set at $0.098 \text{ m}^3/\text{s}$, and in A2 and B2 at $0.267 \text{ m}^3/\text{s}$. Series A and B differed in their initial sorting profiles (Figure 3c). The initial bed of A1 (stage A0) was a flat sloping bed

composed of a mixture of equal proportions of the three size fractions. The initial bed of B1 (stage B0) consisted of a mixed layer of 3 cm on top of a substratum composed of only the finest size fraction. The top layer was composed of the same initial mixture as in A0. Experiments A2 and B2 started from the final stages of A1 and B1, respectively. The initial stage B0 was set up in this way to study the process of (sudden) entrainment of sediment from a fine substratum and the gradual redistribution of size fractions over depth.

2.3. Measurements

[14] This section briefly describes the measuring instruments and the measurements; see also *Blom and Kleinhans* [1999] and *Blom* [2000] for the experiments with the Rhine mixture and the trimodal mixture, respectively.

[15] Longitudinal water and bed level profiles were measured using the standard automatic measuring carriage. It consists of one conductivity sensor for measuring the longitudinal profile of the water level and three conductivity sensors for measuring bed level profiles at three locations over the width of the flume. This type of sensor was developed by WL Delft Hydraulics and measures local electrical resistance or conductivity, which is strongly affected by the presence of water or sand. During a measurement, its vertical position is automatically adjusted by striving after a constant conductivity. In this way, it remains at a fixed position above the interface air-water or water-sediment. The carriage moves in downstream direction with a velocity of about 0.15 m/s, recording water and bed levels every 0.01 m in longitudinal direction. In experiments A1, A2, and B1, longitudinal profiles were measured every 15 minutes, in B2 every 10 minutes, and in the Rhine

A0	A1N	A1E	A1E	A2N	A2E
0	5:57 h	25:01 h	0	4:59 h	18:24 h
B0	B1N	B1E	B1E	B2N	B2E
0	7:04 h	33:21 h	0	3:39 h	24:50 h

Figure 5. Times (in flow hours) of the core sampling sessions in the experiments with the trimodal mixture.

mixture experiments about every hour. Stochastic errors in these measurements are negligible considering that the bed and water level profiles were averaged over the flume width and length and often also over the equilibrium period.

[16] The rate and composition of the total load transport were measured as described earlier. In the experiments with the Rhine mixture, the rate and composition of the bed load and suspended load transport were also measured separately [*Blom and Kleinhans*, 1999; *Kleinhans and Van Rijn*, 2002]. Within the present study, only the measurements of the total load transport are discussed.

[17] Vertical sorting profiles were measured using a new type of core sampling box developed at the University of Twente in 1997. The vertical and two horizontal dimensions of the sampling box were 30 cm, 10 cm, and 15 cm, respectively. To take a core sample, the flume was drained and the sampling box was driven into the bed from a platform. After closing the sampling box by pushing a rectangular plate obliquely into it, the box was removed from the bed (Figure 4). The holes in the bed were filled with sediment layers having a similar composition as removed from the bed. The core sample was cut into thin layers using a thin plate (Figure 4). The thickness of the layers was varied between 3 and 10 mm (potential error 1 mm, but not cumulative over layers), depending on the expected variation in the sorting profile over depth. In the experiments with the Rhine mixture, 11 core samples were taken both at the start and at the final equilibrium stage of each experiment. To study the time evolution of the sorting profile, in the experiments with the trimodal mixture, samples were also taken once before equilibrium was reached (the nonequilibrium N stage) in addition to the initial stage and the equilibrium stage (E stage) (see Figure 5). In each sampling session, 15 core samples were taken. Samples were taken along the flume axis, at the sides, at the bed form crests, in the troughs, as well as at the stoss sides. A remark needs to be made on the dispersive effects of the newly developed coring method, as it may cause relatively coarse grains located near the walls of the sampling box to be pushed down with it. In the present paper no corrections were made for it. As a consequence, in the presented figures the coarse grains may be found at elevations somewhat deeper than their real location. The maximum error in the measured volume fraction content of a certain size fraction at a certain elevation was found to be as large as 0.2, which occurred for the very sharp and unnatural transition in the vertical sorting profile of stage B0. For more natural sorting profiles, the potential error is estimated to be much smaller, about 0.05, especially when the sorting profile is averaged over a number of core samples.

[18] In the experiments with the Rhine mixture, grain size distributions were determined as follows. Samples of the transported sediment and subsamples of the cores were analyzed using a wet sieving method, in which the material

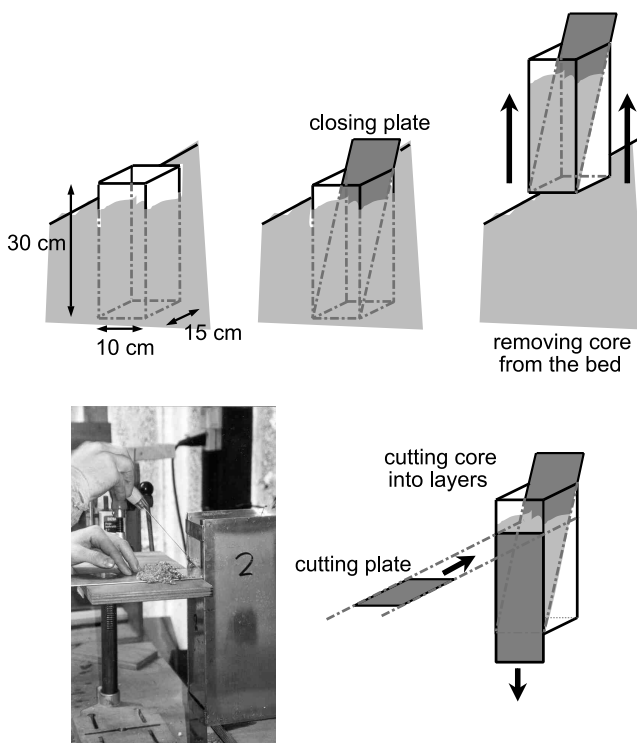


Figure 4. Taking core samples from the bed using a new type of core sampling box (top plots) and cutting the core sample into thin layers (bottom plots).

Table 1. Experimental Parameters (Averaged Over Equilibrium Period)

Experiment	Q_w , m ³ /s	h , m	u , m/s	Fr	i_E , 10 ⁻³	C , m ² /s	R , m	τ , N/m ²
T5 ^a	0.254	0.245	0.69	0.45	1.5	37	0.224	3.4
T7	0.419	0.354	0.79	0.42	1.5	36	0.316	4.8
T9	0.272	0.260	0.70	0.44	1.8	34	0.240	4.1
T10 ^a	0.170	0.193	0.59	0.43	1.2	41	0.176	2.0
A1 ^b	0.098	0.154	0.64	0.52	2.0	38	0.140	2.8
A2	0.267	0.320	0.83	0.47	1.8	38	0.271	4.6
B1 ^c	0.098	0.155	0.63	0.51	1.9	39	0.140	2.6
B2	0.267	0.389	0.69	0.35	2.2	25	0.351	7.4

^aNew fully mixed bed.^bNew bed A0.^cNew bed B0.

was divided into four size fractions: <2 mm, 2–4 mm, 4–8 mm and >8 mm. The smallest size fraction (<2 mm) was analyzed further using a large settling tube, which was calibrated to a standard sieving instrument for a wide range of samples [Blom, 1998]. During the experiments with the trimodal mixture, grain size distributions were determined by wet sieving only, using sieves with sieve diameters of 1.4 mm and 3.35 mm. The wet sieving method leads to a potential error in the measured weight percentage of a certain size fraction of about 3%.

3. Results

3.1. Experimental Parameters

[19] Since for each experiment the initial sorting profile and flow conditions were uniform over the flume length and width, at each point in time all properties of flow, sediment transport, and the vertical sorting profile were uniform over the flume (with the exception of fluctuations due to the bed forms). Table 1 shows the parameters of each experiment, averaged over the flume width, the flume length (reduced by the bed form adaptation length), and over the period in which the bed form dimensions, the bed roughness, and the sediment transport rate and composition varied about a stable mean. From now on, this period is called the equilibrium period. The symbols denote the flow discharge (Q_w), the water depth (h), the average flow velocity (u), the Froude number (Fr), the energy slope (i_E), the Chézy roughness coefficient (C), the hydraulic radius (R), and the bed shear stress (τ). The Chézy roughness coefficient, the hydraulic radius, and the bed shear stress were corrected for sidewall roughness, using the method of Vanoni and Brooks [1957].

3.2. Bed Forms

[20] Figure 6 shows the bed form pattern in the equilibrium stage of experiment T7: coarse particles gather at the trough elevations and the lee faces show a downward coarsening trend. For the experiments with the trimodal mixture, the equilibrium bed form patterns are shown in Figure 7. Barchan-shaped bed forms were present in the equilibrium stage of experiment A1 (phase A1E). The naturally white-colored coarsest size fraction was primarily found in a coarse layer underneath the migrating bed forms. The increase in the discharge in A2 led to more two-dimensional bed forms and an increase in bed form height and

length. Starting from stage B0, experiment B1 showed a pattern of bed forms that was very similar to A1. The fine substratum appeared not to be exposed to the flow significantly, as it did not noticeably affect the pattern of bed forms. After increasing the discharge in B2, the coarse layer broke up and large bed forms developed with ripples on their stoss sides. Ripples were also observed on the stoss sides of the higher bed forms in the Rhine mixture experiments.

[21] Table 2 shows the bed form parameters averaged over the flume width, the flume length (reduced by the bed form adaptation length), and the equilibrium period. In A1 and B1 the bed forms did not migrate over the complete width of the flume (see Figure 7). For these experiments, the parameters were averaged over the adjusted width W_o , which denotes the width of the flume over which the bed forms migrated. The other symbols denote the bed form length (λ), the average crest elevation relative to the average bed level (η_c), the average trough elevation relative to the average bed level (η_b), the average bed form height ($\Delta = \eta_c - \eta_b$), the average bed form shape factor ($\alpha = -\eta_b/\Delta$) defined such that the average bed level η_a is located at $\alpha\Delta$ above the average trough elevation η_b (for triangular dunes $\alpha = \frac{1}{2}$), the average bed form migration speed (c), and standard deviations of the parameters (σ_x). The ripples on the stoss sides of the bed forms in B2 were not included in calculating the average bed form height and bed form length, since in B2 the avalanche mechanism at the lee faces of the large scale bed forms is held responsible for the observed trends in the vertical sorting. The bed forms in experiments T5, T10, A1, and B1 may be defined as Barchan-type bed forms. Note their large ratio of water depth to bed form height, see column h/Δ in Table 2. The bed forms in the other experiments can be defined as dunes.

[22] Figure 8 shows the time evolution of bed form height (top plots) and trough elevation (bottom plots) for experiments A1 and B2 in the flume axis. These two experiments were chosen as they illustrate some important effects of the presence of a coarse layer. First, in A1 the equilibrium bed form height and trough elevation were reached slowly, whereas in B2 they were reached very quickly. In A1, a layer of essentially immobile coarse material quickly formed underneath the small migrating bed forms. The bed form height gradually increased, due to the slow

**Figure 6.** Bed form pattern in equilibrium stage of experiment T7.

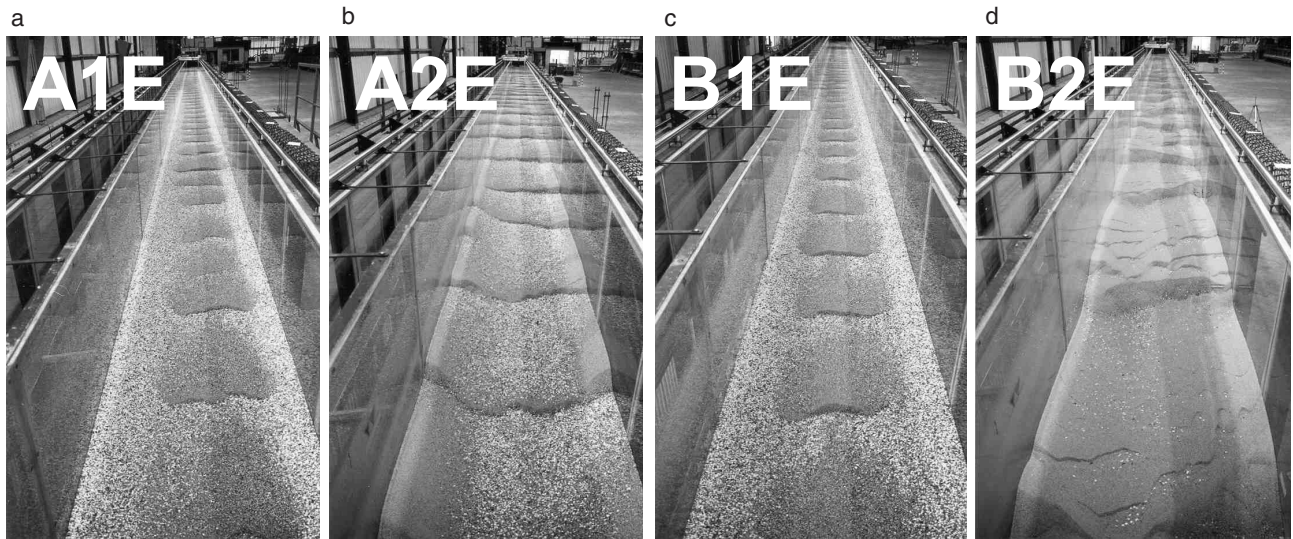


Figure 7. Bed form patterns in equilibrium stages of experiments with trimodal mixture (coarsest fraction is white).

winnowing of fines from the trough surface and subsurface. Experiment B2 started from the final stage of B1 (B1E), in which small bed forms migrated over a coarse bed layer. In B2, this coarse layer quickly broke up and the dune height adjusted to the new flow conditions very quickly. The next section will show that in B2E the transport rate was four times higher than in A1E, which alone cannot explain the smaller adaptation timescales of B2, as the adaptation timescales in B2 were more than four times smaller than in A1. A second difference between experiments A1 and B2 is seen in the time evolution of the trough elevation (Figure 8, bottom plots). After the coarse layer broke up in B2, the troughs reached deep bed elevations and the avalanching process at the dune lee redistributed the medium and coarse size fractions over depth, slightly raising the trough elevation again (note the overshoot of the trough elevation in Figure 8, bottom right plot).

[23] The top plots in Figure 9 show the probability density functions (PDFs) of bed elevations for stages A1E and B2E. This probability density function represents the probability density that the elevation of the bed surface equals bed elevation z , indicating the likelihood of bed elevation z being exposed to the flow. Whereas B2E shows a long tail of deep bed elevations that are exposed to the flow only occasionally, A1E shows a much more abrupt transition at the lower boundary. This is due to the small variability in trough elevations in A1 (Figure 9, bottom plots), which is attributed to the presence of a distinct coarse

layer underneath the migrating bed forms. Also *Wilcock and Southard* [1989] found that the presence of a coarse bed layer underneath migrating bed forms considerably reduces the variability in trough elevations. Hitherto, the interaction between the vertical sorting profile and the probability density function of bed elevations has received little attention and is not well understood. A first and qualitative step was made by *Leclair and Blom* [2001].

3.3. Sediment Transport

[24] Table 3 shows the total load transport rates averaged over the equilibrium period, in which Q_{tot} denotes the volume of total load transport per unit time and q_{tot} the volume of total load transport per unit (adjusted) width and time. For the experiments with the trimodal mixture, Table 3 also shows the volume fraction contents of the fine, medium, and coarse size fractions in the total load transport (F_1 , F_2 , and F_3 , respectively) averaged over the equilibrium periods, together with their standard deviations. The measured total load transport rates showed a natural fluctuation between roughly 0 and $2Q_{tot}$, which was due to the natural variation in the transport rate over the bed forms.

[25] Figure 10 shows the composition of the total load in the experiments with the Rhine mixture. In each experiment the transported material was much finer than the original bed material, indicating partial transport. Differences in transport composition between the experiments were small. The number of samples of the transported

Table 2. Bed Form Parameters Averaged Over Equilibrium Period and (Adjusted) Flume Width

Experiment	W , m	W_o , m	λ , m	η_b , cm	η_{bs} , cm	Δ , cm	α	c , m/h	h/Δ	σ_{λ} , m	σ_{η_b} , cm	$\sigma_{\eta_{bs}}$, cm	σ_c , m/h
T5	1.5	-	1.13	2.1	-1.7	3.8	0.45	-	6.4	0.34	1.0	0.7	-
T7	1.5	-	1.53	3.4	-3.2	6.6	0.49	-	5.4	0.42	1.3	1.2	-
T9	1.5	-	1.34	2.8	-2.9	5.7	0.51	-	4.5	0.39	1.1	1.1	-
T10	1.5	-	0.79	1.0	-0.7	1.7	0.41	-	11.3	0.42	0.6	0.3	-
A1	1.0	0.68	0.90	0.8	-0.8	1.7	0.50	10.3	9.3	0.28	0.6	0.4	0.6
A2	1.0	-	1.38	2.0	-2.9	4.9	0.59	8.8	6.5	0.39	1.2	0.9	1.2
B1	1.0	0.65	0.99	0.9	-0.9	1.8	0.49	10.9	8.5	0.25	0.6	0.4	0.8
B2	1.0	-	1.79	5.0	-7.3	12.2	0.59	3.7	3.2	0.61	2.6	2.6	0.6

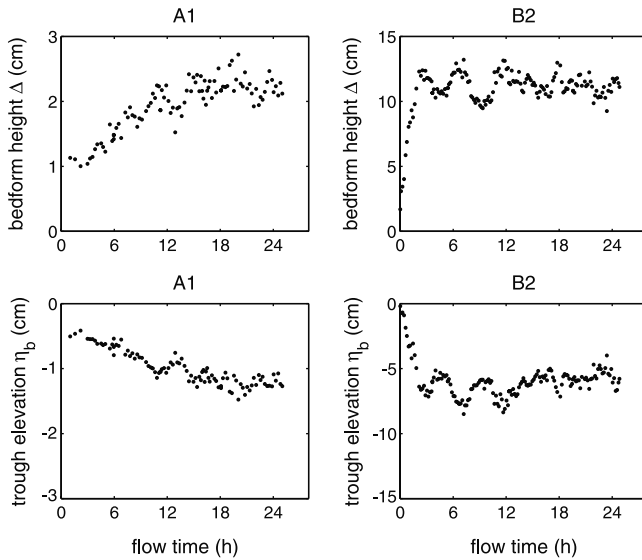


Figure 8. Time evolution of bed form height (top plots) and trough elevation (bottom plots) in the flume axis for experiments A1 and B2. Note that the vertical scale for B2 is 5 times larger than for A1.

material was limited though, so that no definite conclusions are drawn.

[26] In the experiments with the trimodal mixture, equilibrium in the rate of the total load transport was reached in 2 to 5 flow hours. In all experiments but B2, equilibrium in its composition was reached after only 2 flow hours (Figure 11). The breaking up of the coarse layer explains the initially large amount of the medium and coarse size fractions in the total load. After this, material from the fine substratum became available to the transport process. Surprisingly, in A1 and B1 the volume fraction content of the medium size fraction in the total load was larger than the finest size fraction. This was attributed to an extreme form of hiding of

Table 3. Parameters of Total Load Transport, Averaged Over Equilibrium Period

Experiment	Q_{tot} $10^{-5} \text{ m}^3/\text{s}$	q_{tot} $10^{-5} \text{ m}^2/\text{s}$	F_1	F_2	F_3	σ_{F_1}	σ_{F_2}	σ_{F_3}
T5	4.73	3.15						
T7	8.19	5.46						
T9	5.66	3.77						
T10	1.72	1.15						
A1	1.13	1.67	0.32	0.51	0.17	0.14	0.09	0.07
A2	4.33	4.33	0.38	0.38	0.24	0.11	0.09	0.10
B1	1.17	1.79	0.34	0.54	0.12	0.11	0.09	0.04
B2	4.45	4.45	0.90	0.05	0.05	0.09	0.05	0.05

the finest size fraction in the pores of the coarse bed layer. This seems to be caused by the rather unnatural grain size distribution of the well sorted trimodal mixture, having two distinct ranges of “missing” grain sizes.

[27] For the experiments with the trimodal mixture, the measured transport rates are compared with the transport rates resulting from the method of dune tracking. This could not be done for the experiments with the Rhine mixture, as bed form migration speeds could not be determined with sufficient accuracy. The dune tracking method is based on the simple-wave approach, in which bed forms are assumed to migrate (1) without a change in dune shape, (2) with a constant migration speed, and (3) with all grains participating in the avalanching process at the lee face, and (4) without sediment fluxes between subsequent dunes [Bagnold, 1941]. The transport rate then equals:

$$q_a = cc_b \alpha \Delta \quad (1)$$

where q_a denotes the volume of bed load transport per unit width and time, and c_b denotes the sediment concentration within the bed ($c_b = 1 - \lambda_p$, where λ_p denotes the porosity). Figure 12 shows a comparison between the measured transport rates and the transport rates based on this simple wave approach. The values agree very well, which indicates that (1) the amount of suspended load was indeed negligible and (2) the amount of particles by-passing the lee face and thus not participating in the avalanching process [Mohrig and

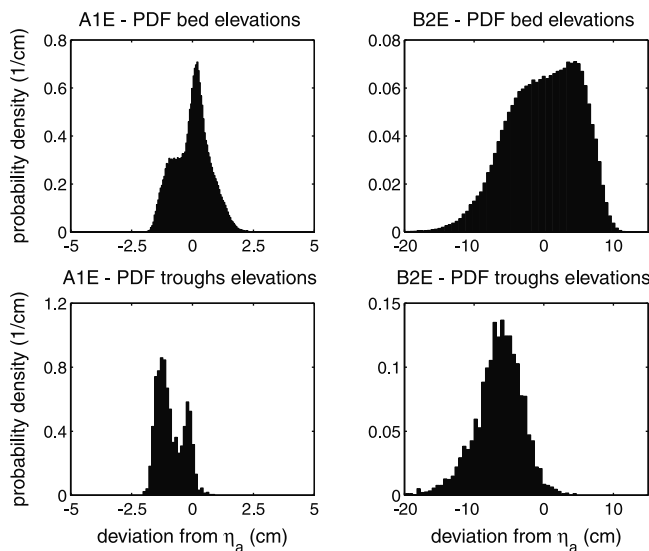


Figure 9. Probability density functions (PDFs) of bed elevations (upper plots) and of trough elevations (lower plots), for the equilibrium periods of experiments A1 and B2.

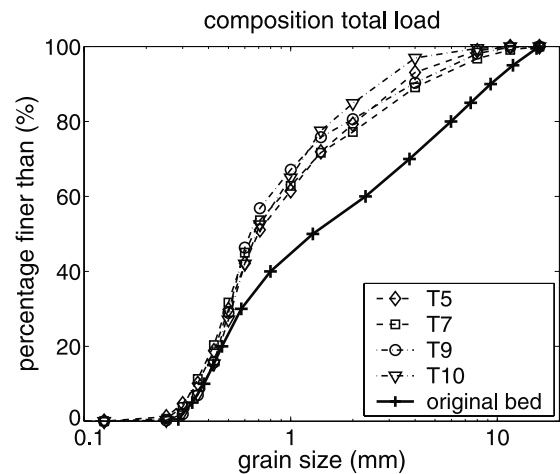


Figure 10. Composition of total load transport in experiments with the Rhine mixture (averaged over equilibrium period).

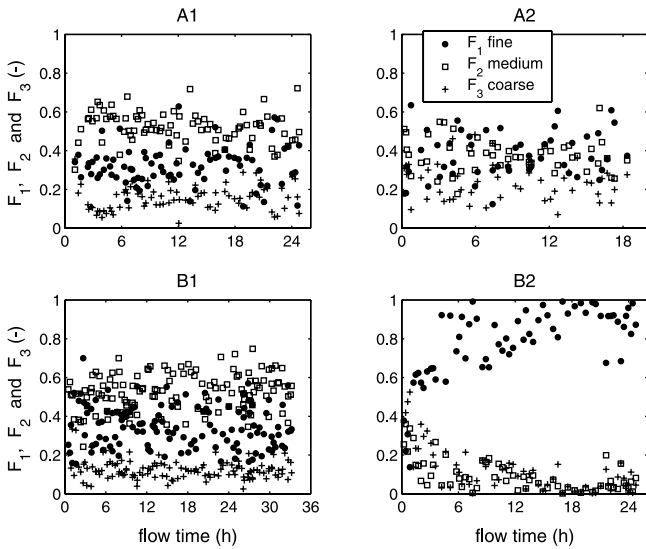


Figure 11. Time evolution of the composition of the total load in the experiments with the trimodal mixture. Legend is given in the top right plot.

Smith, 1996] was negligible, as well. In the experiments with the Rhine mixture, measurements of suspended load concentrations indicated that the amount of suspended load transport varied between 10 and 20% of the total load, depending on the flow conditions [M. G. Kleinhans, Utrecht University, personal communication, 2000].

3.4. Vertical Sorting Profiles

[28] Figure 13 shows the results of six core samples taken at the final stages of experiments T5, T7, and T9, which together resemble a flood hydrograph (Figure 3). For each of these experiments, Figure 13 shows both one sample taken at or near a bed form crest (left-hand plots) and one sample taken in a trough (right-hand plots). The plots show the variation of D_{10} , D_{30} , D_{50} , D_{70} , and D_{90} of the bed material over depth, as well as the D_{10} to D_{90} of the original mixture.

[29] When comparing the variation in D_{50} over depth with the D_{50} of the original mixture for the crest sample of

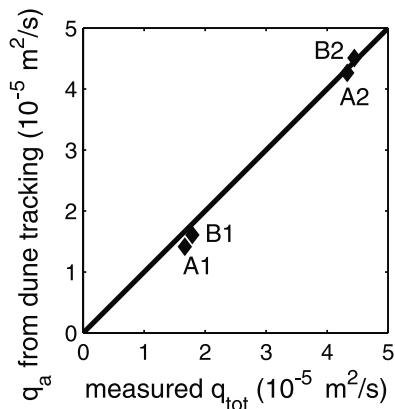


Figure 12. Measured total load transport q_{tot} compared with bed load transport rates based on the simple-wave approach q_a for the experiments with the trimodal mixture. Parameters are averaged over equilibrium period.

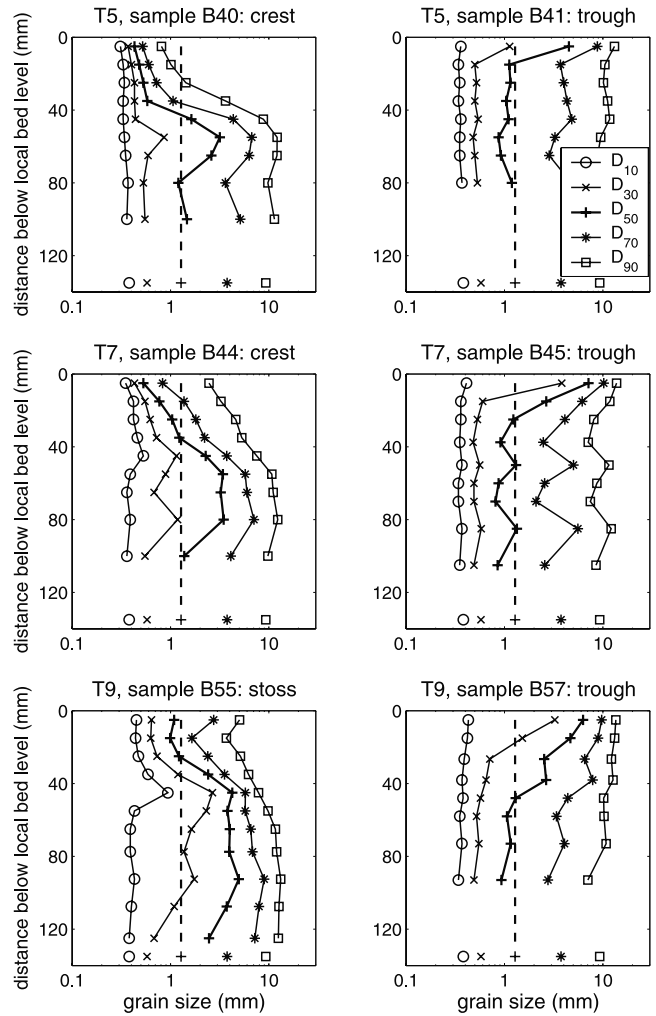


Figure 13. Core samples taken at crests and in troughs in experiments T5, T7, and T9 (Rhine mixture): variation of D_{10} , D_{30} , D_{50} , D_{70} , and D_{90} over depth (see legend in top right plot) compared with D_{10} , D_{30} , D_{50} , D_{70} , and D_{90} of the original mixture (see symbols at bottom of each plot). The D_{50} of the original mixture is also indicated by the vertical dashed line.

T5 (upper left-hand plot), we recognize the downward coarsening of the dune material. Between 4 and 8 cm below the local bed level, the bed material was coarser than the original mixture, indicating the presence of a coarse bed layer. In the crest sample of T7 (middle left-hand plot), we again recognize the downward coarsening of the bed form material and the presence of a coarse bed layer, which is now located further below the local bed level than in T5, coinciding with the larger bed forms in T7. The sample in T9 was taken at the stoss side (lower left-hand plot) and shows the downward coarsening of the dune material and the presence of a now very thick coarse layer.

[30] Comparing the crest and trough samples of each experiment, the trough surface appears to be coarser than the coarse layer underneath the dunes. This is attributed to the winnowing of fines from the trough surface and subsurface, although the mechanism was not directly measured. The winnowing led to a higher porosity of the trough surface and a coarsening of it. As soon as a bed form

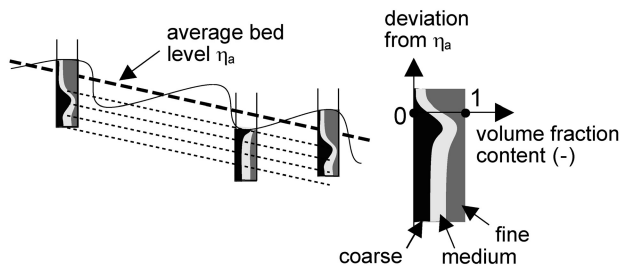


Figure 14. Method of averaging core samples in experiments with the trimodal mixture.

migrated over this area, the pores were filled again with finer material and the coarse layer became “less coarse.”

[31] Figure 14 shows the method to determine the average vertical sorting profile, i.e., the variation of the volume fraction contents of size fractions over depth. This was only done for the experiments with the trimodal mixture. The vertical sorting profiles were averaged along lines parallel to the average bed level profile. This was done for all stages, i.e. the initial stage, the nonequilibrium N stage, and the equilibrium E stage. For each stage, also the probability distribution of bed elevations was determined, so that the vertical sorting in the bed can be related to the likelihood of a certain bed elevation being exposed to the flow.

[32] Figure 15 shows the time evolution of the vertical sorting profile for experiment A2, for core samples taken only at or near bed form crests. It shows the sorting profiles of the initial stage of A2 (phase A1E, Figure 15a), the nonequilibrium stage, after 5 flow hours (phase A2N, Figure 15b), and the equilibrium stage, after 18.5 flow hours (phase A2E, Figure 15c) (see also Figure 5). Figure 7a showed that in phase A1E small bed forms migrated over a coarse bed layer. In Figure 15a, we recognize the downward coarsening within the bed forms. However, a coarse layer underneath the bed forms cannot be identified. This will be clarified in one of the next paragraphs.

[33] At the top of the sorting profiles in Figure 15, we distinguish a top layer of 1 to 1.5 cm that is coarser than the

bed form material just below. This seems to have two reasons. First, for taking the core samples, the flow needs to be turned off, and the sediment that is being transported over the stoss face settles on top of it. This sediment is coarser than the bed form material. Second, processes of selective entrainment act on the stoss face. Fine grains are entrained and transported more easily than the coarser ones, which may result in a slight coarsening of the surface. The top layer seems to have a thickness larger than a few grain sizes, yet, this is due to the averaging over a large number of samples. The core samples were taken at different elevations relative to the average bed level, and at each elevation the dune surface is covered by this coarse top layer. The presence of such a coarse top layer on the stoss face was also observed by Ribberink [1987].

[34] In the equilibrium stage of A2 (phase A2E), the dune height had increased and more coarse material was present in the active part of the bed (Figure 15c). A coarse layer was located at about 30 mm below the average bed level. For the nonequilibrium stage of A2 (phase A2N), the amount of coarse material in the upper part of the bed was larger than in A2E and a coarse layer was found at about 25 mm below the average bed level (Figure 15b). This indicates that, after 5 flow hours, the coarse material was still being reworked to deeper bed elevations.

[35] Let us now come back to the reason for the “absence” of the coarse layer in Figure 15a. Figure 16 presents the average vertical sorting profiles for phase A1E averaged over core samples taken at or near bed form crests (Figure 16a) and in troughs (Figure 16b). The trough surface at 20 mm below the average bed level was significantly coarser than the bed material underneath the small bed forms at that same elevation. Turbulent forces in the trough area caused finer particles to be winnowed from the trough surface and subsurface. When a bed form migrates over this area, the pores in the coarse layer become filled again with fines. As shown in Figure 16a, this process even caused the coarse layer to be “absent” in the vertical sorting profile averaged over only crest samples.

[36] Figure 17 presents the average vertical sorting profiles in phase B0, the equilibrium stage of B1 (phase B1E),

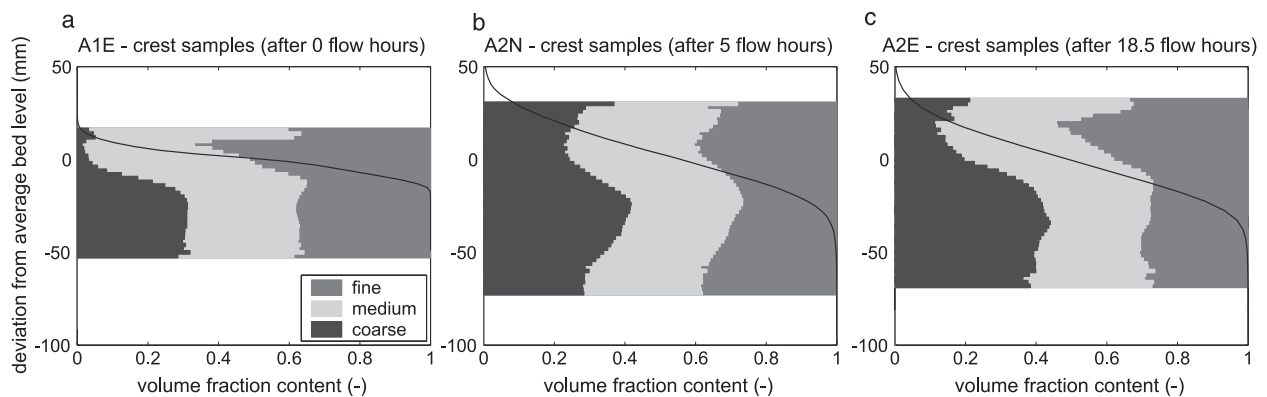


Figure 15. Experiment A2: time evolution of the vertical sorting profile averaged over samples taken at or near bed form crests in (a) the initial stage, after 0 flow hours (phase A1E), (b) the nonequilibrium stage, after 5 flow hours (phase A2N), and (c) the equilibrium stage, after 18.5 flow hours (phase A2E). The solid line represents the probability distribution of bed elevations in the corresponding phase of the experiment.

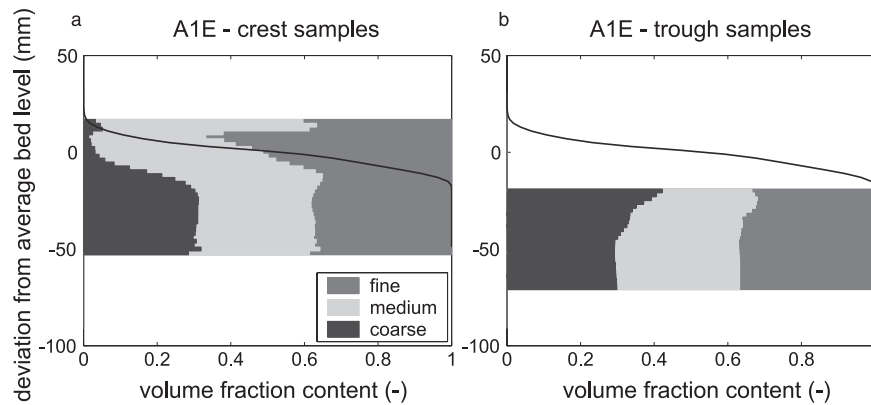


Figure 16. Equilibrium stage of experiment A1 (phase A1E): vertical sorting profiles averaged over (a) samples taken at or near bed form crests and (b) samples taken in troughs. The solid line represents the probability distribution of bed elevations in the corresponding phase of the experiment.

and the equilibrium stage of B2 (phase B2E). The original interface between the two layers is indicated by the dashed line at 30 mm below the average bed level. However, Figure 17a shows that in the initial stage B0, the medium and coarse size fractions were also found below this interface. This was due to the coring method: driving the sampling box into the bed caused coarser particles that were located very close to the walls of the box to be pushed down with it. This dispersive effect was not corrected for. As a consequence, in the figures presented here, the coarse particles may be found at elevations somewhat deeper than their real location. An alternative may be to use the non-destructive sampling method developed by *Marion and Fraccarollo* [1997], in which large elements of the bed are frozen, removed, analyzed, and brought back into the flume.

[37] We may conclude that the fine substratum did not have much influence on B1 (Figure 17b), since experiment B1 was very similar to A1 (with respect to bed load transport rate and composition, bed form height, bed roughness, and bed surface composition). In B2 the coarse layer quickly broke up and the medium and coarse size fractions were gradually redistributed over a large depth. In B2 it was observed that in the avalanching process the medium and

coarse size fractions were often overtopped by other (finer) grains before they reached the trough elevations.

4. Discussion

[38] In all experiments, the proportion of coarse fractions in the bed load transport was smaller than in the substratum, which indicates that partial transport conditions were present. The study was aimed at conditions with bed load transport only, which were met very well in the experiments with the trimodal mixture (Figure 12). However, in the experiments with the Rhine mixture, the amount of suspended load varied between 10 and 20% of the total load transport. In the present study we do not consider the impacts of suspended load on the vertical exchange of sediment in the bed.

[39] Figure 18a shows an interpretation of experiments T5, T7, and T9. In experiment T5, which started with a fully mixed bed, small barchan-type bed forms developed that migrated over a coarse bed layer. In experiment T7, the flow velocity was increased, which resulted in higher dunes and a lowering of the coarse layer. This was probably caused by a combination of (1) the (partial) breaking up of the coarse layer at the start of T7, after which large bed forms

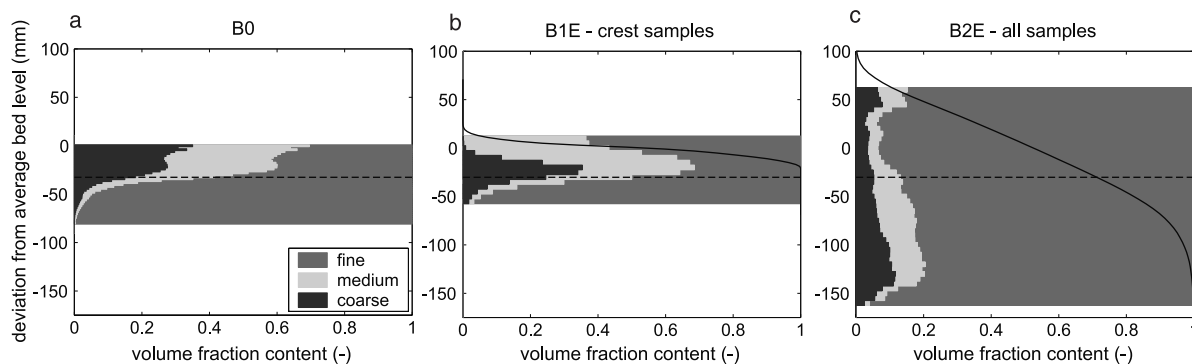


Figure 17. Vertical sorting profiles in (a) initial stage (phase B0), averaged over all samples (flat bed), (b) equilibrium stage of experiment B1 (phase B1E), averaged over only crest samples, and (c) equilibrium stage of experiment B2 (phase B2E), averaged over all samples. The solid line represents the probability distribution of bed elevations in the corresponding phase of the experiment. The dashed line represents the original interface (in B0) between the top layer and the fine substratum.

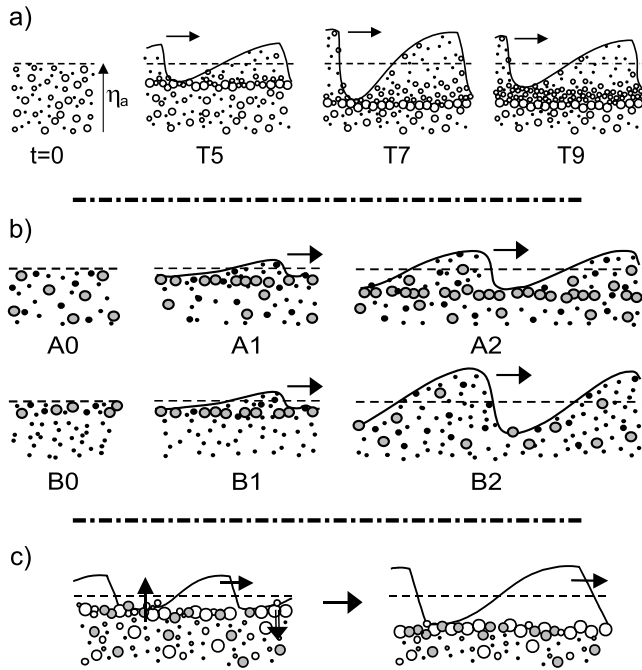


Figure 18. Interpretation of (a) equilibrium stages of experiments T5, T7, and T9, (b) equilibrium stages of experiments with the trimodal mixture, and (c) winnowing of fines from the trough surface and subsurface resulting in a lowering of the coarse layer. For simplicity, the variation in trough elevations is not drawn. For a better schematization of the sorting trend within the bed forms, see Figure 1.

developed and the coarse particles were deposited at the lower elevations through the avalanche process at the lee face, and (2) the winnowing of fines from the trough surface and subsurface. In experiment T9, the flow velocity was decreased to the same level as in T5. Particles that were transported in experiment T7, but could no longer be transported in T9, settled on top of the coarse layer that remained as a relic at its deep elevation. The dune height decreased but remained higher than in T5, since more fines had been brought above the level of the protecting coarse layer during T7. In a similar set of experiments, *Klaassen* [1992] did not find significant changes in the vertical sorting between two experiments with similar flow conditions before and after a flood discharge. This, however, may be due to *Klaassen* taking bed samples only at dune crests where the coarse layer may not be identified very well due to the filling up of its pores underneath the migrating dunes.

[40] Figure 18b shows an interpretation of the equilibrium stages of the experiments with the trimodal mixture. In experiment A1, a layer of coarse material quickly formed underneath the migrating bed forms. The bed form height increased gradually through the slow process of winnowing and the lowering of the coarse layer. The presence of the coarse layer limited the variability in trough elevations. After increasing the flow in A2, the dune height increased and the coarse layer was found at a deeper elevation, which was due to the combination of the avalanching process at the dune lee and the winnowing of fines, similar to T7. Experiment B1 was very similar to A1, so that we may

conclude that the fine substratum did not have much influence on the final stage of B1.

[41] An especially interesting case is experiment B2, in which the adaptation timescale of the vertical sorting was of the same order as the adaptation timescale of the bed load transport composition. Due to the quick breaking up of the initially present coarse layer, underlying fine material became available and large dunes developed with deep and irregular trough elevations. The avalanching process at the dune lee slowly redistributed the coarse particles over depth, slightly raising the trough elevation again.

[42] A mechanism that may have a significant effect on the vertical sorting in dunes is the winnowing or washing out of fines from the trough surface and subsurface [see also *Wilcock and Southard*, 1989; *Klaassen*, 1990]. The mechanism of winnowing has never been directly measured and modeled, yet it is attributed to the character of the statistics of the highly turbulent flow in the trough areas. *McLean et al.* [1994] studied the flow structure over dunes and emphasized the important effects of the statistics of the turbulent flow in the trough areas upon sediment entrainment. Winnowing occurs if (1) partial transport conditions prevail and a relatively large amount of coarse material does not participate in the transport process, (2) the framework of coarse particles is permeable enough to let finer particles pass, and (3) the statistics of the highly turbulent flow in the trough areas allows fine material to be washed out.

[43] The winnowing of fines causes the porosity of the trough surface and subsurface to increase and the coarse layer to coarsen. Gradually, more bed material becomes available to the transport process and the coarse layer may eventually settle (Figure 18c). As soon as a dune migrates over the trough area, the pores get filled again with finer material. This may even cause the coarse layer underneath migrating bed forms not to be recognized. The winnowing and lowering of the coarse layer is expected to continue until (1) the sediment transport rate equals the sediment transport capacity or (2) enough coarse material is gathered to form a stable armor layer through which no sediment can be exchanged (in situations with a large amount of coarse bed material) or (3) the water depth becomes so large that the turbulence intensity in the troughs becomes too small to entrain bed material.

[44] For flows with bed shear stresses that are well above the threshold for incipient motion of all size fractions, it is expected that the coarse bed layer will break up and participate in the transport process [see also *Wilcock and Southard*, 1989; *Klaassen*, 1990]. The breaking up of the coarse layer is initiated by the highly turbulent flow in the trough areas. Yet, it is expected that the trend of downward coarsening within the dunes maintains but becomes less pronounced, as the mechanism of grain size selective deposition over the lee face seems only slightly dependent on the bed shear stress. This situation of having only mobile size fractions and thus mobile coarse sediment in the lower parts of the bed forms was studied by *Ribberink* [1987].

[45] Experiment A1 showed a very slow adaptation of the average trough elevation, whereas B2 shows an initially very quick and large adaptation and then a second adaptation period in which the adaptation is much slower and smaller (see Figure 8). Although vertical sorting profiles were measured at only three points in time during an

experiment, the authors believe that the adaptation timescale of the vertical sorting is strongly related to the adaptation timescale of the trough elevation and to the variability in trough elevations, as, through migration, a dune redistributes all bed material above its trough elevation. It is important to note that the adaptation timescale of the vertical sorting depends on the bed elevation. First, a deeper elevation of the active bed is less exposed to the flow, so that entrainment of particles from and deposition to this elevation occur less frequently. Secondly, at a deeper elevation of the active bed more sediment is present. Thus a higher rate of sediment exchange is required to reach the same amount of change in composition. Both factors cause the timescale of reaching the equilibrium sorting profile to be larger for deeper bed elevations, and need to be taken into account when modeling the vertical redistribution of sediment within the bed.

[46] The results of the present study show that vertical sediment fluxes are not restricted to conditions with net degradation or aggradation and that these fluxes also occur through the migration of bed forms and through the variability in trough elevations. Nonetheless, in most sediment continuity models for nonuniform sediment, vertical sediment fluxes only occur in situations with net degradation or aggradation. At present, the results presented here are being used for deriving formulations for the vertical exchange fluxes in the Parker-Paola-Leclair framework for sediment continuity. A first step toward the modeling of the vertical sorting processes of nonuniform sediment was the modeling of the vertical fluxes of tracer particles in uniform sediment in dune conditions [Blom *et al.*, 2001].

5. Conclusions

[47] Two new data sets were developed describing size selective transport and vertical sorting in conditions with bed forms and bed load transport. These data sets will be very useful in modeling vertical sorting processes within the river bed and the formation of coarse bed layers.

[48] Vertical sediment fluxes within the bed are not restricted to situations with net aggradation or degradation but also occur through the migration of bed forms and through the variability in trough elevations. The results presented here indicate that the description of vertical sorting processes in existing sediment continuity models for nonuniform sediment, e.g. the Hirano active layer model, requires modification to account for sorting other than through of net aggradation or degradation.

[49] Three mechanisms contribute to the formation of an essentially immobile coarse layer underneath migrating dunes: (1) the avalanching process at the lee face, (2) partial transport in which a relatively large amount of coarse material does not participate in the transport process, and (3) the winnowing of fines from the trough surface and subsurface.

[50] After a flood event, a coarse bed layer may be found at the trough elevations accompanying the highest dunes (compare sorting profiles in experiments T5, T7, and T9, Figure 18a). During lower flows with smaller dunes, this layer of coarse material may act as a relic, no longer taking part in the transport process.

[51] The winnowing of fines from the trough surface and subsurface was also recognized by Wilcock and Southard

[1989] and Klaassen [1990], but it has never been directly measured. It is attributed to the character of the statistics of the highly turbulent flow in the reattachment zone. The winnowing of fines may cause (1) the trough surface and subsurface to have an increased porosity and to coarsen, (2) more bed material to become available to the transport process, and (3) the coarse layer to settle until a new dynamic equilibrium between entrainment and deposition in the troughs is reached. When a dune migrates over the trough area, the pores become filled again with finer particles. By taking core samples at dune crests only, one may not recognize the presence of a coarse layer underneath the dunes.

[52] The results of the present study point toward a close relation between (the adaptation timescale of and the variability in) the trough elevations and (the adaptation timescale of) the vertical sorting. This can easily be understood when considering that, through migration, a dune redistributes all bed material above its trough elevation. Within a series of dunes, the variability in trough elevations can be large, also in conditions without net degradation or aggradation. The variability in trough elevations is strongly limited when a distinct coarse layer is present underneath migrating dunes.

[53] It is important to note that whether the lower (coarser) part of a bed form is mobile or essentially immobile cannot be concluded from the vertical sorting profile itself. This can only be assessed when studying the sorting profile in combination with the probability distribution of bed elevations. Additionally, it would be helpful to study the vertical sorting profiles at locations of both bed form crests and troughs.

Notation

c	bed form migration speed, m/s.
c_b	sediment concentration within the bed, dimensionless.
C	Chézy roughness (corrected for sidewall roughness according to Vanoni and Brooks), $m^{1/2}/s$.
F_i	volume fraction content of size fraction i (1 fine, 2 medium, 3 coarse) in the total load transport, dimensionless.
Fr	Froude number, dimensionless.
h	water depth, m.
i_E	energy slope, dimensionless.
q_a	volume of bed load transport per unit (adjusted) width and time, m^2/s .
q_{tot}	volume of total load transport per unit (adjusted) width and time, m^2/s .
Q_{tot}	volume of total load transport per unit time, m^3/s .
Q_w	flow discharge, m^3/s .
R	hydraulic radius (corrected for sidewall roughness according to Vanoni and Brooks), m.
u	average flow velocity, m/s.
W	flume width, m.
W_o	flume width reduced by width over which hardly any sediment transport occurs, m.
α	average bed form shape factor, dimensionless.
Δ	bed form height, m.
η_a	average bed level, m.
η_r	crest elevation relative to average bed level, m.
η_b	trough elevation relative to average bed level, m.
λ	bed form length, m.

- λ_p porosity, dimensionless.
 σ_g geometric standard deviation of sediment mixture, dimensionless.
 σ_x standard deviation of parameter x , unity as for x .
 τ bed shear stress (corrected for sidewall roughness according to Vanoni and Brooks), N/m^2 .

[54] **Acknowledgments.** The experiments with the Rhine mixture were funded by the European Union as a Training and Mobility of Researchers (TMR) project and by the combination RIZA-WL-UT (the Institute for Inland Water Management and Waste Water Treatment of the Ministry of Transport, Public Works, and Water Management in the Netherlands, WL Delft Hydraulics, and the University of Twente). In 1997, these three institutes started a joint project to finance the Ph.D. research of the first author. The experiments with the Rhine mixture were conducted by Astrid Blom and Jan Ribberink (University of Twente), Maarten Kleinhans and Janrik van den Berg (University of Utrecht), Klaus Basso and Andreas Dittich (University of Karlsruhe), and Luigi Fraccarollo and Aronne Armanini (University of Trento). The authors especially thank Maarten Kleinhans for his participation. The experiments with the trimodal mixture were funded by the combination RIZA-WL-UT and conducted by Astrid Blom, Peter van der Scheer, and Jan Ribberink. The authors thank Joop Ouderling, Freek de Groot, and Erik Mosselman of WL Delft Hydraulics for their assistance during the experiments. Suzanne Leclair and the reviewers are thanked for their valuable comments on the paper.

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